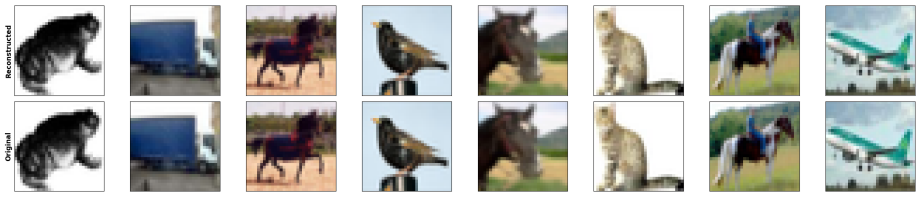
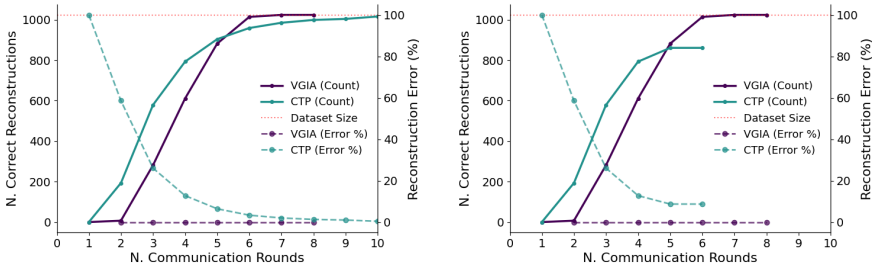
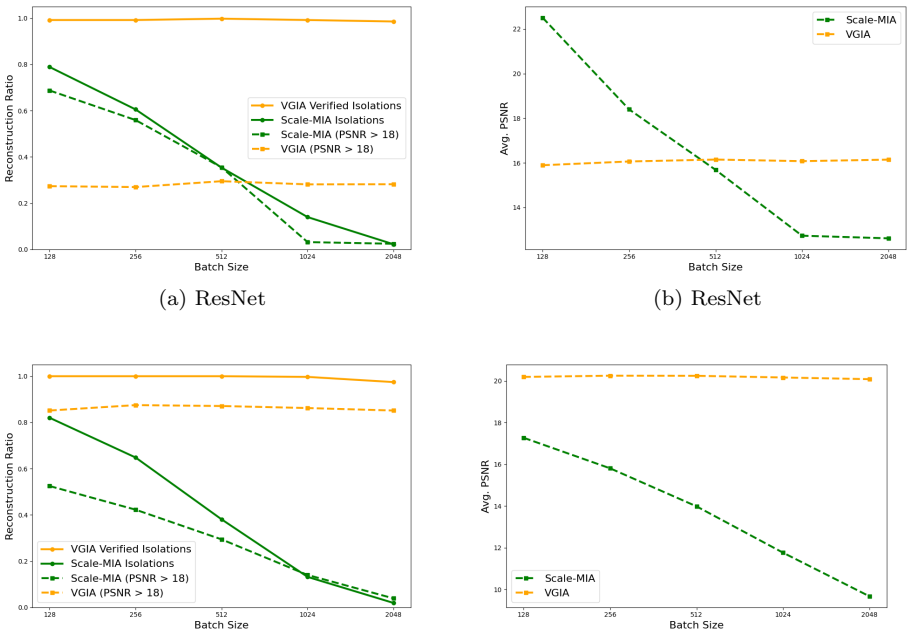


Figure 1: Examples of reconstructions (top) from VGIA on CIFAR-10 using a three-layer fully connected neural network with batch size $B = 1024$



(a) VGIA vs. $\text{CTP}_{=}$ (b) VGIA vs. $\text{CTP}_{>}$

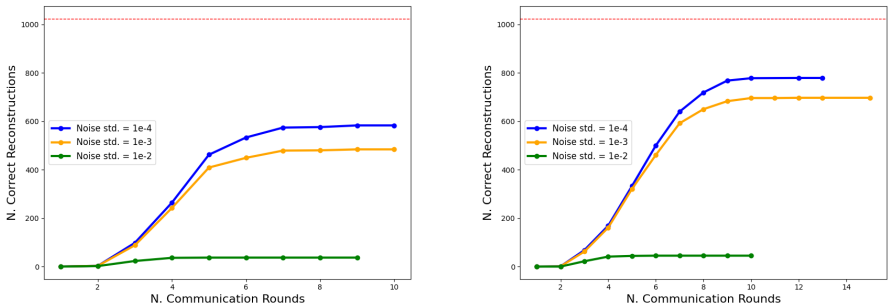
Figure 2: Attack performance of VGIA and the CTP baselines when the target client trains a three-layer fully connected neural network on a random subset of ImageNet (size 1024) using full-batch FedSGD. (a,b) Number of correct reconstructions and reconstruction errors under varying attack-round budgets. $\text{CTP}_{=}$ presents a baseline with perfect knowledge of $\varepsilon_{\mathbf{w}} =$ (the minimum absolute distance between the projections of any two samples along direction $\mathbf{w}$). $\text{CTP}_{>}$ presents a too optimistic ε, i.e., $\varepsilon = 10^{-2} > \varepsilon_{\mathbf{w}}$.



(a) ResNet (b) ResNet

(c) Transformer (d) Transformer

Figure 3: Comparison of VGIA with Scale-MIA (Shi et al. 2025) on ResNet and Visual Transformer (ViT) architectures under varying batch sizes. For both models, the attack layer contains $N = 512$ neurons. (a,c) Fraction of samples correctly isolated between two consecutive hyperplanes and fraction of images reconstructed with PSNR > 18 under different batch sizes. Note that VGIA can verify isolation through Eq. (12), while Scale-MIA has no tool to check it. (b,d) Average PSNR across different batch sizes. A decoder specifically trained for VGIA could further improve its performance.



(a) HARUS (b) CIFAR-10

Figure 4: Number of reconstructed inputs from a batch of size $B = 1024$ using a three-layer fully connected neural network, under different levels of gaussian noise with variance σ^2. The attack layer has $N = 1000$ neurons. (a) Results on HARUS dataset. (b) Results on CIFAR-10.

[0.0000,0.0000,0.0000,0.0000,0.0000,0.0000,0.5190,0.7826,0.3980,1.0000]

(a)

[0.0000,0.0000,0.0000,0.0000,0.0000,0.0000,0.5190,0.7826,0.3980,1.0000]

(b)

Figure 5: Example of (a) an input and (b) its VGIA reconstruction from a batch of $B = 2048$ samples from the ACS Income dataset. The last 10 feature values are shown.

[0.1818, 0.5938, 0.4136, 0.0098, 0.4000, 0.0000, 0.7500, 0.5000, 0.9167,
0.4616,0.2635,0.8522,0.0000,0.0253,0.6358,0.3397,0.7006,0.0162]

(a)

[0.1818, 0.5938, 0.4136, 0.0098, 0.4000, 0.0000, 0.7500, 0.5000, 0.9167,
0.4616,0.2635,0.8522,0.0000,0.0253,0.6358,0.3397,0.7006,0.0162]

(b)

Figure 6: Example of (a) an input and (b) its VGIA reconstruction from a batch of $B = 2048$ samples from the King County dataset.

$$[0.6176, -0.0081, -0.0658, 0.7867, 0.4193, 0.3887, 0.7803, 0.3772, 0.3688, 0.9080]$$

(a)

$$[0.6176, -0.0081, -0.0658, 0.7867, 0.4193, 0.3887, 0.7803, 0.3772, 0.3688, 0.9080]$$

(b)

Figure 7: Example of (a) an input and (b) its VGIA reconstruction from a batch of $B = 2048$ samples from the HARUS dataset. The first 10 feature values are shown.

$$[0.2393, -0.0243, -0.0959, -0.9904, -0.9930, -0.9937, -0.9909, -0.9932, -0.9957, -0.9408]$$

(a)

$$[0.2684, -0.0200, -0.1046, -0.9904, -0.9880, -0.9903, -0.9906, -0.9874, -0.9901, -0.9392]$$

(b)

Figure 8: Example of (a) a correct reconstruction by VGIA and (b) an incorrect reconstruction by $\text{CTP}_{>}$, for a batch of $B = 2048$ samples from the HARUS dataset. The first 10 feature values are shown. The incorrect reconstruction corresponds to a convex combination of original samples, with all features lying within the input bounds $[-1, 1]$. As a result, it is not possible to visually distinguish correct from incorrect reconstructions.

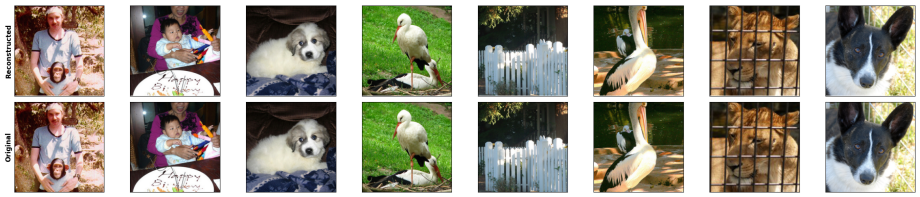


Figure 9: Examples of inputs (bottom) and their VGIA reconstructions (top) when training on ImageNet with a three-layer fully connected neural network with $B = 1024$.

Figure 10: Examples of inputs (bottom) and their VGIA reconstructions (top) on CIFAR-10 for a ResNet model from a batch of $B = 1024$ samples.

Figure 11: Examples of inputs (bottom) and their VGIA reconstructions (top) on CIFAR-10 for a ViT model from a batch of $B = 1024$ samples.

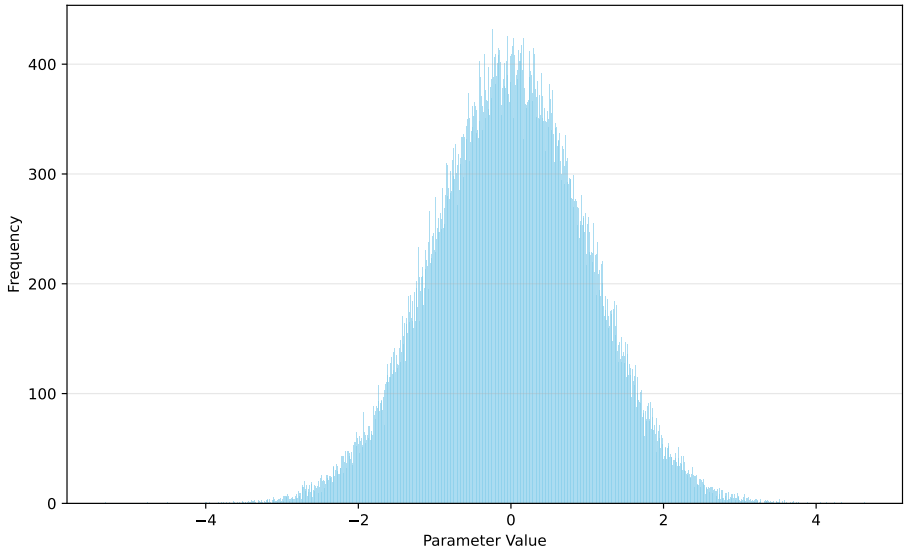


Figure 12: Distribution of the malicious parameters initialized from a normal distribution $\mathcal{N}(0, 1)$. Results are on a three-layers fully connected NN with $b = 2048$ on ACS Income dataset.

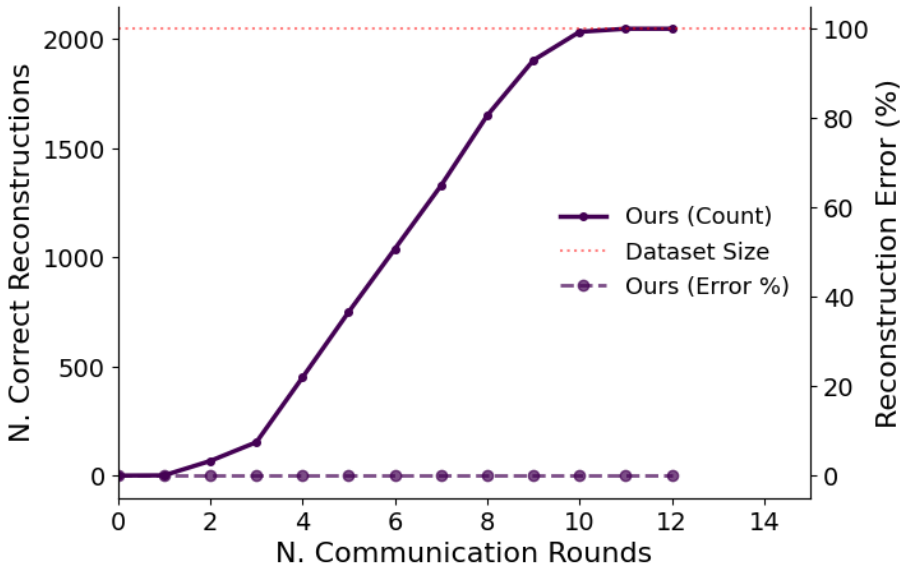


Figure 13: VGIA performance on the ACS Income dataset when parameters of intermediate layers $(2 : L)$ are initialized from $\mathcal{N}(0, 1)$. Results are obtained using a three-layer fully connected neural network with $b = 2048$.